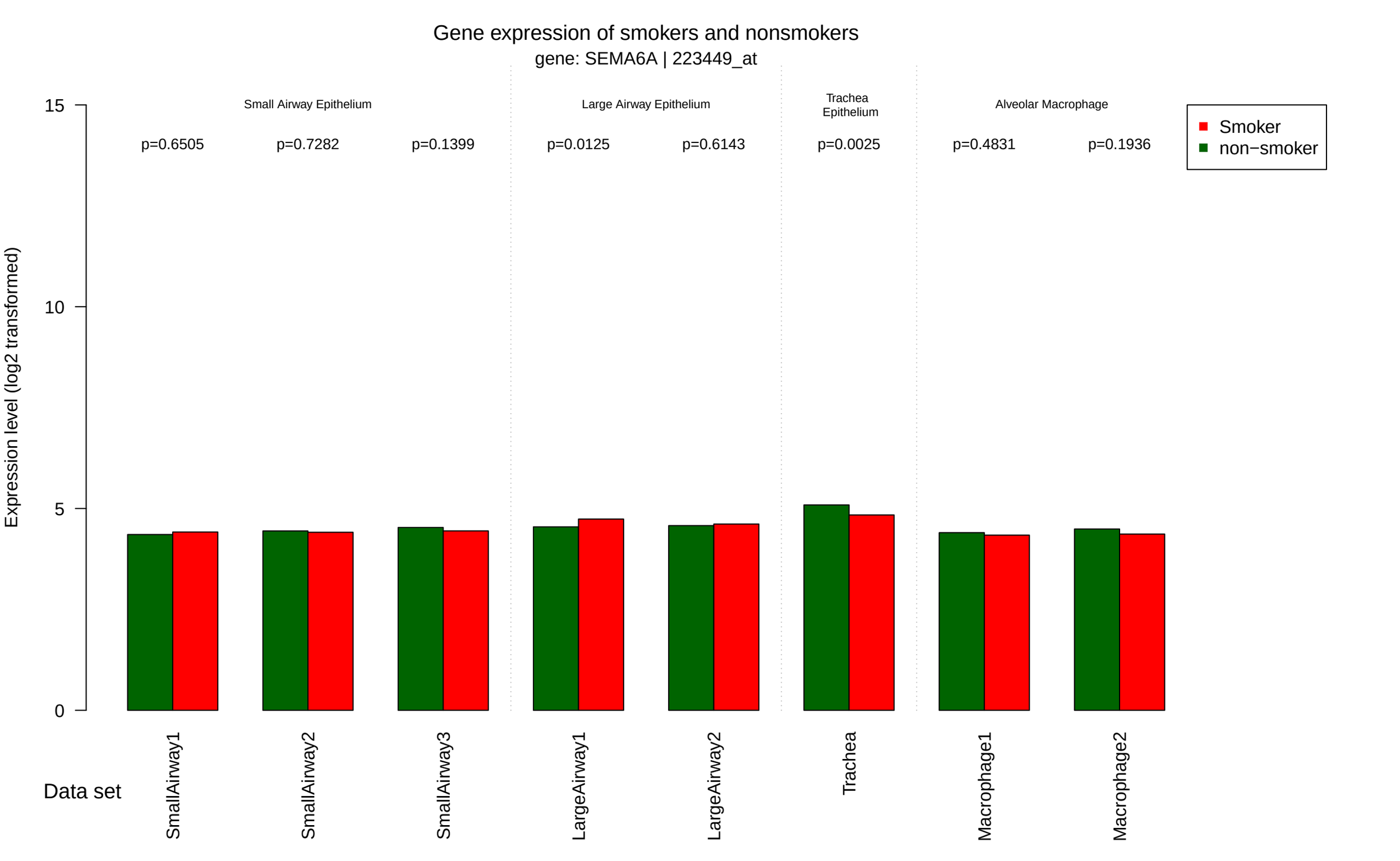
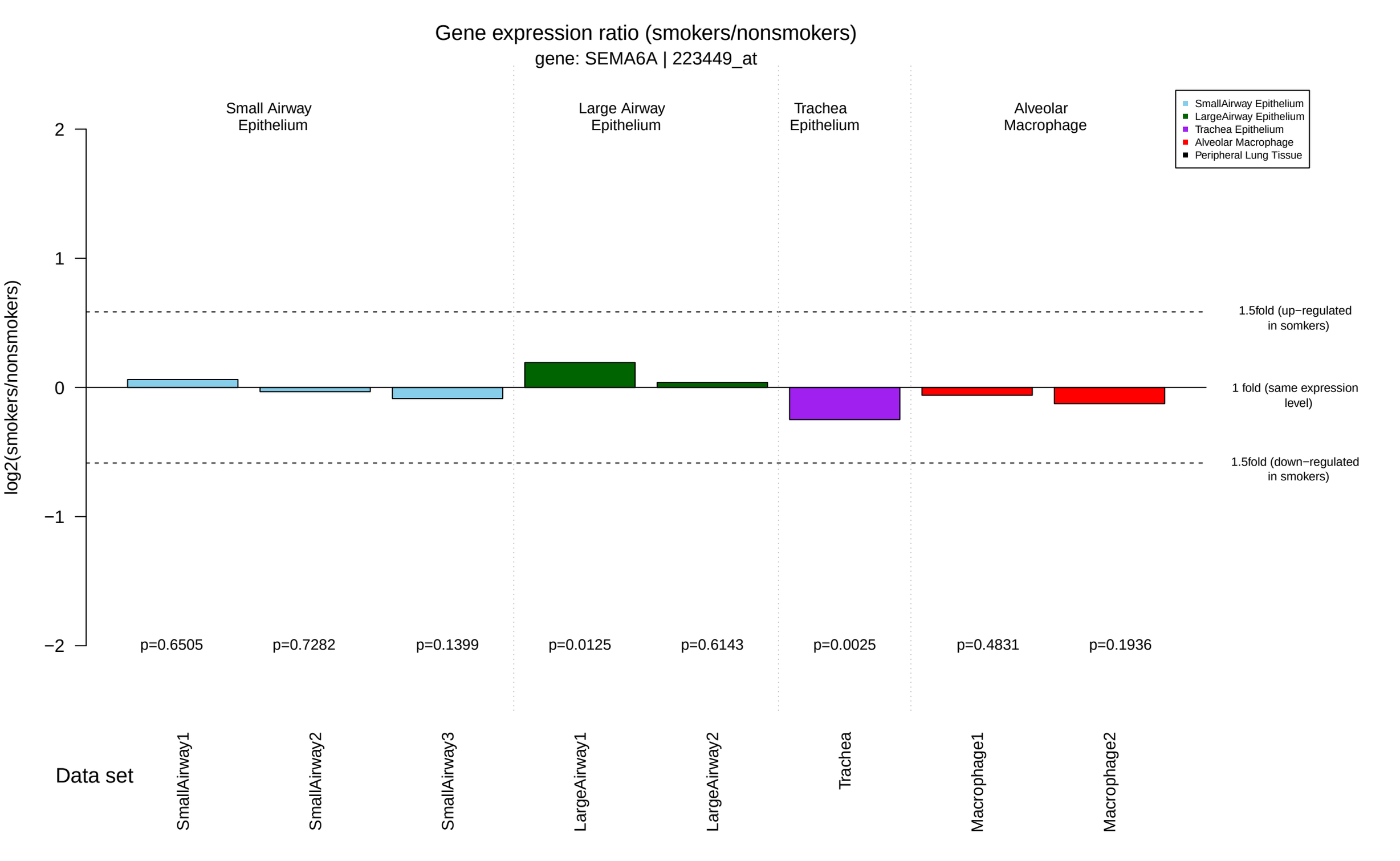
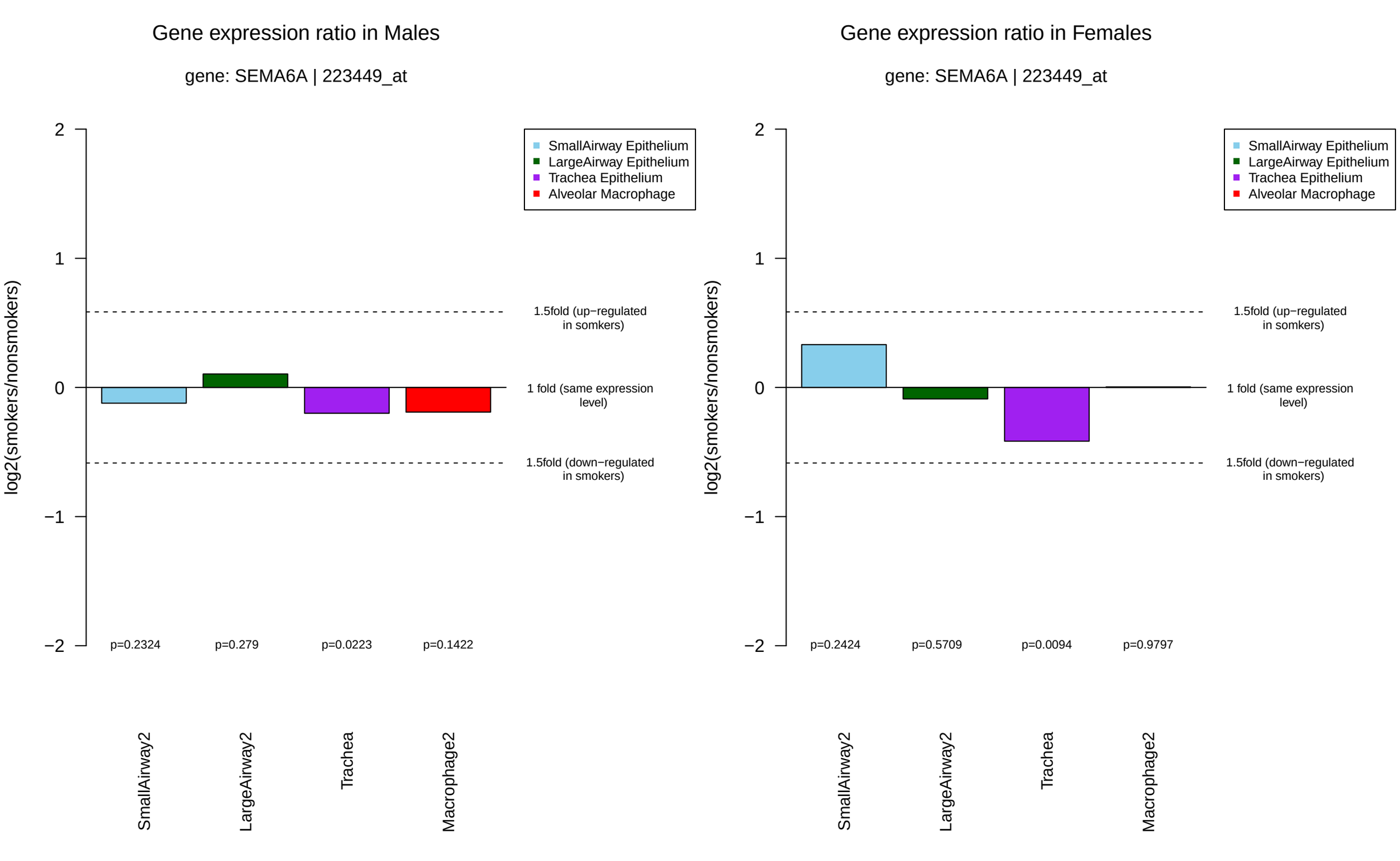
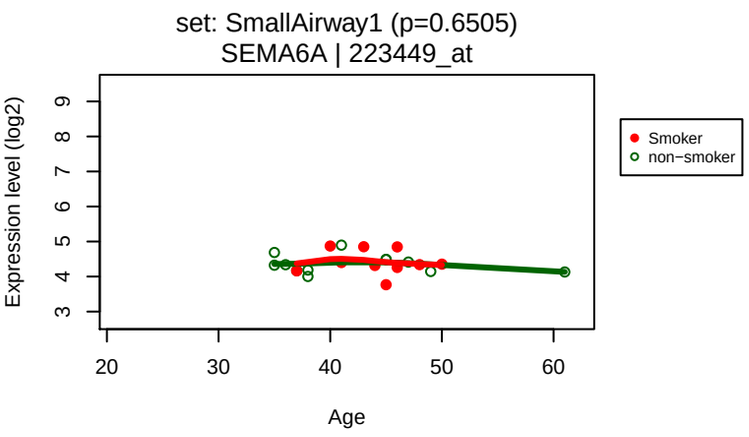
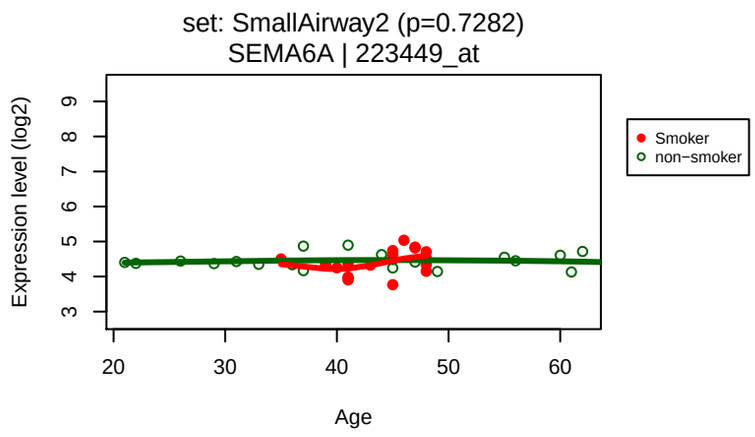
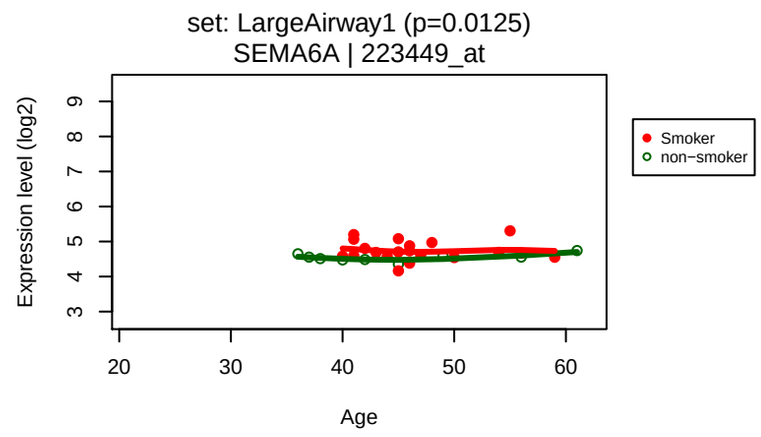

Gene expression of smokers and nonsmokers
gene: SEMA6A | 223449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6505
p=0.7282
p=0.1399
p=0.0125
p=0.6143
p=0.0025
p=0.4831
p=0.1936
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

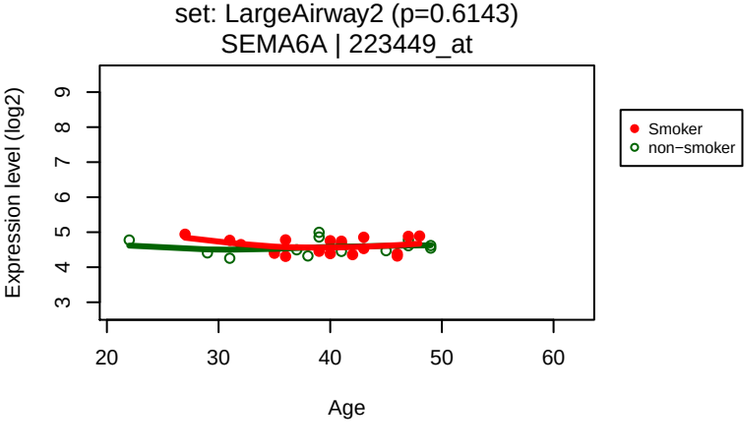

Gene expression ratio (smokers/nonsmokers)
gene: SEMA6A | 223449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6505
p=0.7282
p=0.1399
p=0.0125
p=0.6143
p=0.0025
p=0.4831
p=0.1936
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

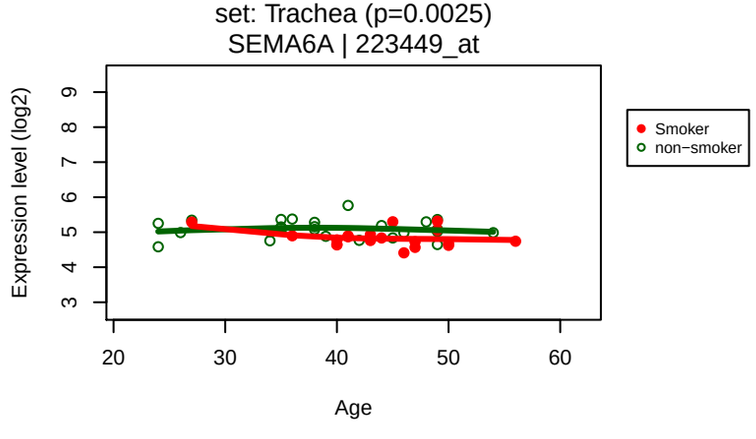

Gene expression ratio in Males
gene: SEMA6A | 223449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2324
p=0.279
p=0.0223
p=0.1422
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SEMA6A | 223449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2424
p=0.5709
p=0.0094
p=0.9797
SmallAirway2
LargeAirway2
Trachea
Macrophage2

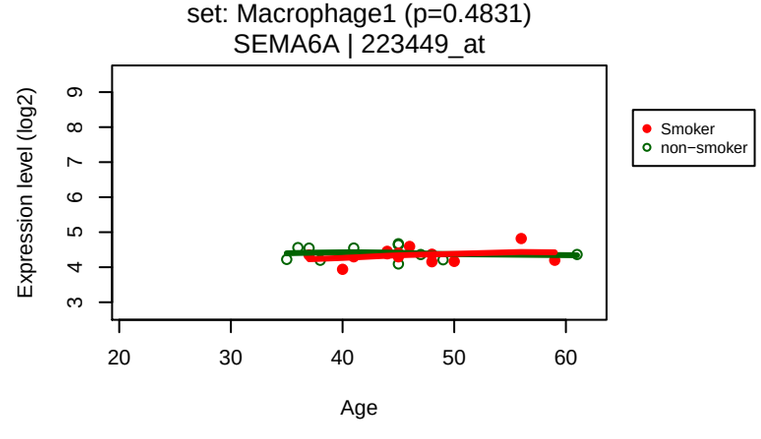

set: SmallAirway1 (p=0.6505)
SEMA6A | 223449_at
Expression level (log2)
Age
Smoker
non−smoker

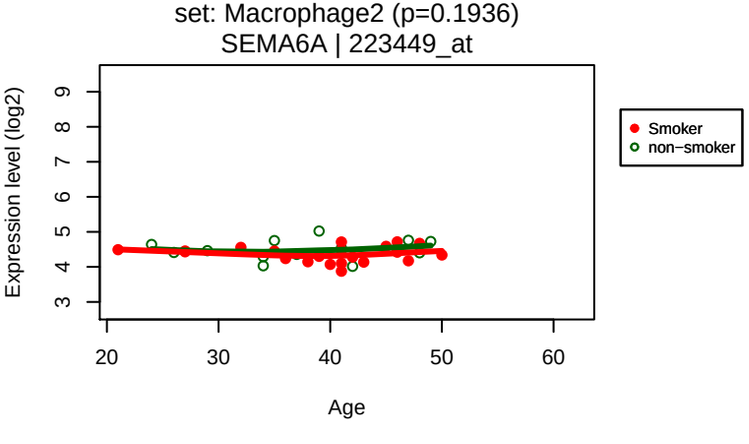

set: SmallAirway2 (p=0.7282)
SEMA6A | 223449_at
Expression level (log2)
Age
Smoker
non−smoker

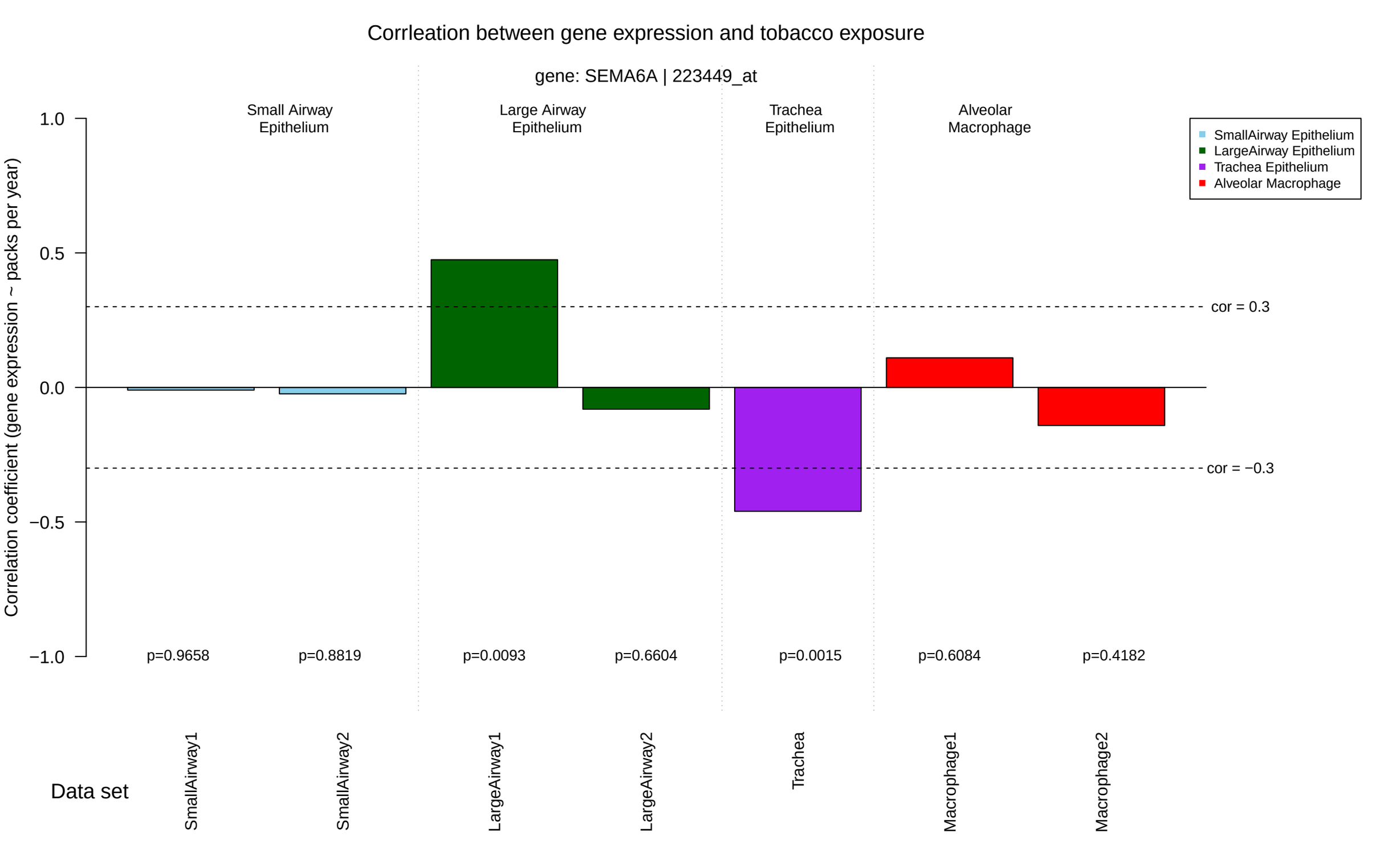

set: LargeAirway1 (p=0.0125)
SEMA6A | 223449_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6143)
SEMA6A | 223449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0025)
SEMA6A | 223449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4831)
SEMA6A | 223449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1936)
SEMA6A | 223449_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SEMA6A | 223449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9658
p=0.8819
p=0.0093
p=0.6604
p=0.0015
p=0.6084
p=0.4182
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2